

JERRY NG YIN WEI

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EDUCATION

Bachelor of Engineering (Hons) Industrial & Systems Engineering **Aug 2024 - Present**
National University of Singapore (NUS), Singapore

- NUS Merit Scholarship
- Current CAP: 4.65/5.0

WORK AND INTERNSHIP EXPERIENCES

Simulation Engineering Intern **May 2026 – Aug 2026**
DSO National Laboratories *Singapore*

- Developed Python/FreeCAD automation pipelines to generate parametric engineering models, reducing manual model setup time from ~30 minutes per model to under 2 minutes
- Treated layout generation and model cleaning as constrained optimisation/validation problems; implemented geometric checks to resolve room/wall intersections, improve spatial consistency, and scale scenario generation.
- Implemented CUDA-parallelised Monte Carlo simulations to model projectile trajectory and impact behaviour across generated environments, producing risk-weighted outputs for engineering decision analysis.

Company Quartermaster Sergeant **Feb 2022 - Feb 2024**
Singapore Armed Forces (SAF) *Singapore*

- Managed \$10M+ inventory across units; built Excel asset & lifecycle tracking to streamline key supply workflows and improve audit-ready records.
- Analyzed maintenance and inventory datasets to optimize resource allocation, cut downtime and sharpen demand forecasting for procurement and scheduling.

Junior Marketing Coordinator **Oct 2021 - Jan 2022**
STERIS Life Sciences *Singapore*

- Analyzed competitor intel to shape strategy and executed regional campaigns; delivering a complete product-launch package and two promotional flyers.

SKILLS

Programming: Python, Java, C, SQL, HTML, React
Analytics & Optimisation: Linear Programming, Mixed-Integer Programming, Monte Carlo Simulation, Data Analysis
Tools & Frameworks: FastAPI, AWS Bedrock, Microsoft Excel (Advanced), SAP, Git

ACTIVITIES

SimplifyNext Agentic AI Hackathon — Team Member **Aug 2025 - Sep 2025**

- Built SkillPilot: SG SkillsFuture course recommender (React/FastAPI; Bedrock-Claude; SSG-WSG OAuth2) with NLP resume parsing + cosine similarity to rank courses & sequence paths.

NUS ISE Business Analytics Case Challenge (BACC) — Team Member **Mar 2025**

- Optimized fab capacity planning: matched wafer starts to demand via LP/MIP under capacity & smoothing constraints, improving line loading and cutting backlog risk.
- Ran Monte Carlo on cycle times to size tooling/CAPEX; delivered risk-weighted investment plan.

HTX Investigators' Challenge 2021 — Team Member **Dec 2021**

- Performed cryptanalysis, reverse engineering, and Python scripting to decrypt messages and uncover vulnerabilities.

ACHIEVEMENTS

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- BACC (NUS ISE × Micron) — 3rd Runner-Up, ~100 teams (2024)
 - HTX Investigators' Challenge: Exploiter Award for cryptography (2021)